COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 031 - Visualizations and Data Analysis	**SECTION:**	CPE21S1
DATE PERFORMED:	Aug 8, 2026	**DATE SUBMITTED:**	Aug 8, 2026
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ACTIVITY NO. 4 **Data Analytics Career**

1. PROCEDURE OUTPUT

Provide a lab report on the assigned module from the Introduction to Data Science Course and go to Module 4.1.5 Practice Item - Job Market and 4.2.2 Practice Item - The Importance of Project Portfolios for the Hands-on Activity.

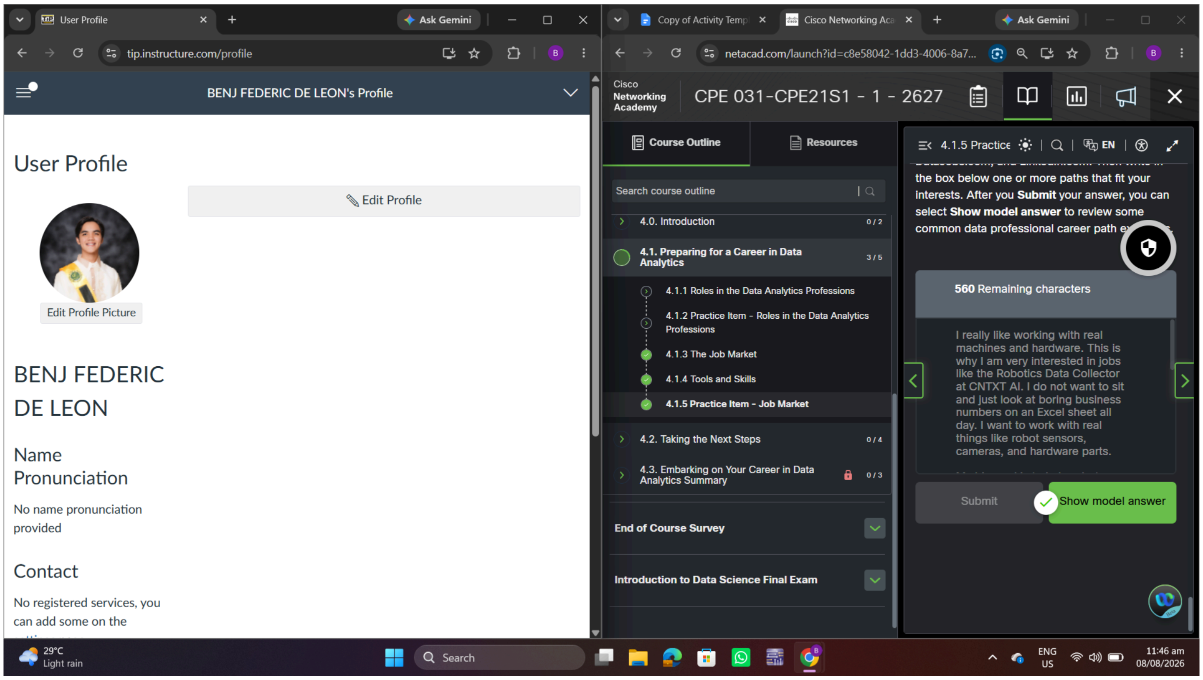
4.1.5

I really like working with real machines and hardware. This is why I am very interested in jobs like the Robotics Data Collector at CNTXT AI. I do not want to sit and just look at boring business numbers on an Excel sheet all day. I want to work with real things like robot sensors, cameras, and hardware parts.

My big goal is to help robots see and understand the world. I want to learn how to collect the data from these machines and make sure the sensors are working good. When a robot moves or makes a mistake, I want to check the logs and find out why the hardware did that. It is very exciting for me to see how computer data makes a physical machine move in real life.

I know I still have things to learn, but I am ready to work hard. I want to be the person who checks the cameras and fixes the data problems at the source. This path is perfect for me because I love robots and I want to help build smart machines for the future.

Every day, robots get more popular in factories and warehouses. I want to be there to handle the physical equipment myself. Connecting the wires, setting up the digital cameras, and testing the laser sensors is the kind of job I enjoy doing. I do not mind getting my hands dirty to fix a mechanical problem. I want to learn how to manage big fleets of machines and track their battery life, speed, and error rates. If a sensor gets dusty or breaks down during a test run, I want to be the one who goes there to clean it, calibrate it, and reset the system. This hands-on work is what makes me happy, and I know I can help the engineering team a lot by making sure the data they get from the hardware is always clean and perfect.



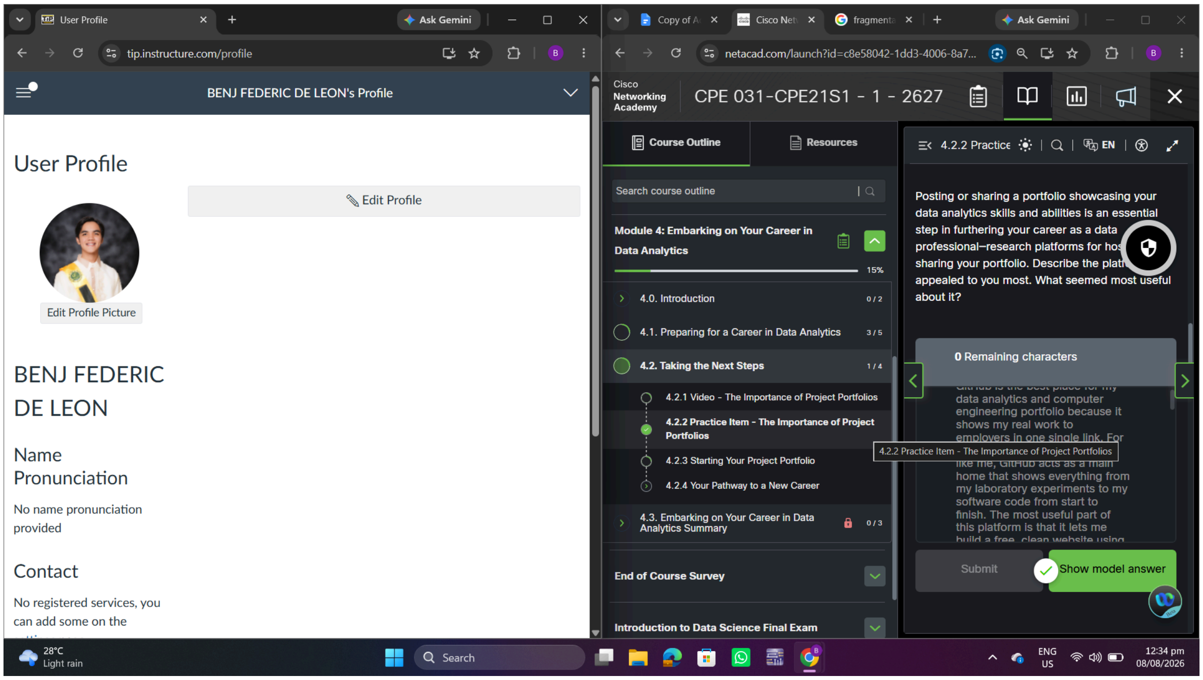
4.2.2

GitHub is the best place for my data analytics and computer engineering portfolio because it shows my real work to employers in one single link. For a second-year BSCpE student like me, GitHub acts as a main home that shows everything from my laboratory experiments to my software code from start to finish. The most useful part of this platform is that it lets me build a free, clean website using GitHub Pages without paying any money. Bosses and hiring managers can click on my project to see my real SQL queries, C++ firmware, and Python automation scripts to see exactly how I clean sensor data and solve hard technical problems. It also shows my circuit diagrams, data charts, and written notes directly on the screen so people do not need to download big files to see what I did.

To fully appreciate why GitHub stands out as the ultimate ecosystem for engineering students, one must look closely at how we execute our academic and personal projects. A traditional engineering or data portfolio often suffers from extreme fragmentation. A typical hiring manager might look at an isolated interactive dashboard or a physical prototype but remain completely blind to the messy, intricate microcontroller programming and data cleansing processes that had to happen beforehand. They might browse through a random code file on a school drive but completely miss out on how that code actually translates into a working, hardware-integrated system. GitHub bridges this massive structural gap by serving as a comprehensive narrative engine for my computer engineering and data analysis pipelines. It successfully transforms abstract technical resume terms, like microcontrollers, IoT, and data structures, into a concrete, interactive, and fully auditable reality.

When a technical recruiter or engineering lead lands on a well-structured GitHub profile, they are not simply scanning a static student resume; they are stepping into my active digital laboratory. Every repository functions as an independent, deep-dive project case study. Through the strategic use of highly optimized README files, I can seamlessly guide an employer through the exact multi-step lifecycle of an engineering problem. This comprehensive journey includes outlining the initial project requirements, hardware components used, exploratory data analysis, sensor calibration, and final performance metrics. By hosting the underlying Python code, Arduino or C++ scripts, and raw sensor datasets in one unified ecosystem, I provide complete technical transparency.

Furthermore, GitHub Pages elevates this professional experience by providing a premium, zero-cost web hosting framework. This allows savvy BSCpE students to deploy customized web applications and beautiful interactive portfolios for recruiters globally. Beyond just hosting raw code, it allows me to build full-scale documentation sites that act as an interactive frontend for my hardware and data solutions. Hiring managers can see embedded metrics, read engineering methodologies, and review system architectures directly from a web browser. This unique combination of code version control, public documentation, and web hosting makes it irreplaceable for computer engineering students trying to break into the competitive tech landscape. This is crucial now as I prepare for upcoming internships and future engineering roles.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: ______________________________

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: ______________________________

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: ______________________________

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.